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BE (8th Semester)

Examination, April - May, 2013

Branch : Mechanical

ROBOTICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. First part of each question is compulsory and attempt any two from the rest. Give neat and clean diagrams whatever needed.

Q. 1. (a) What is meant by the two terms "Robot" and "Robotics" ? 2

(b) How can classify the robots based on the arm geometry ? Explain with neat sketch. 7

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(c) Define 'degree of freedom'. How many degrees of freedom can a wrist have ? What is the purpose of these degrees of freedom in the wrist. 7

(d) Study the human arm anatomy and describe the features a humanoid robot should have. A robot is required to perform an assembly of a shaft into a bearing placed in an arbitrary position. How many degrees of freedom are required for a manipulator to perform this task ? If the bearing is placed in a fixed plane, say a horizontal plane, what will be the required number of degrees of freedom ? Explain. 7

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2. (a) Draw the figure of the end-effector coordinating frame clearly indicating approach, orientation and normal directions and roll, pitch and yaw motions. 2

(b) What are fundamental rotation matrices ? Obtain the three fundamental rotations matrices for rotation about x, y and z axis from the rotation matrix for rotation about an arbitrary axis k. 7

(c) A 2-D rigid motion takes the points (0, 1) and (1, 1) to $\left(\frac{1-\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ and $\left(\frac{2-\sqrt{3}}{2}, \frac{1}{2}\right)$ respectively. Find the 3×3 matrix which effects this transformation and also find its centre of rotation. 7

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(d) Why is it important to choose a frame assignment for n-DOF manipulator that gives a maximum number of zero joint-link parameters ? What is the representation of the reflection R of E^3 into itself, with respect to the x – y plane, in terms of unit vectors parallel to the x, y, z axes that form a coordinate frame F ? 7

Q. 3. (a) What are the two basic parameters for a link for kinematic modelling ? Briefly explain. 2

(b) Explain / write the various steps of the algorithm for link-frame assignment on the basis of DH convention. 7

(c) Draw a schematic diagram for given RRP manipulator, with the axes of frames {0}

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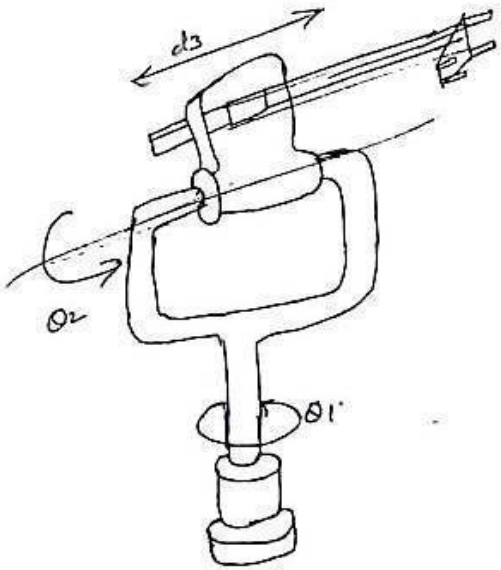
through {3} labeled. Also, include the parameters μ_1, μ_2, a_2 and d_3 on your schematic. Assume that in this diagram, the slider bar is parallel to the ground and that this is the configuration where $\mu_1 = 0$, $\mu_2 = 90°$. Write down the Denavit-Hartenberg parameters for this manipulator, in the form of a table. 7



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(d) Discuss the mechanical structure and notations for modelling of robots. 7

Q. 4. (a) What are the basic guide-lines for choosing the right sensor ? 2

(b) What is 'Segmentation' ? What are the three common techniques for it ? Explain precisely giving suitable diagrams. 7

(c) What is potentio meter ? Explain theory of operation of digital potentiometer. 7

(d) Why do we process images ? What is monochrome image and digital image ? 7

Q. 5. (a) What are the four D's of Robotics which are concerned with safety to human beings ? 2

(b) Discuss the robotic applications in the field of quality control-inspection and testing 7

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(c) How can you justify industrial robots are better for competitive global environment ? 7

(d) What are the non industrial applications of the robot ? Explain at least five non industrial applications of robot. 7

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